Microelectronics Circuit

Evolution of Electronic Devices

Vacuum Tubes

(a)

(b)

Discrete Transistors

Integrated Circuits

(c)

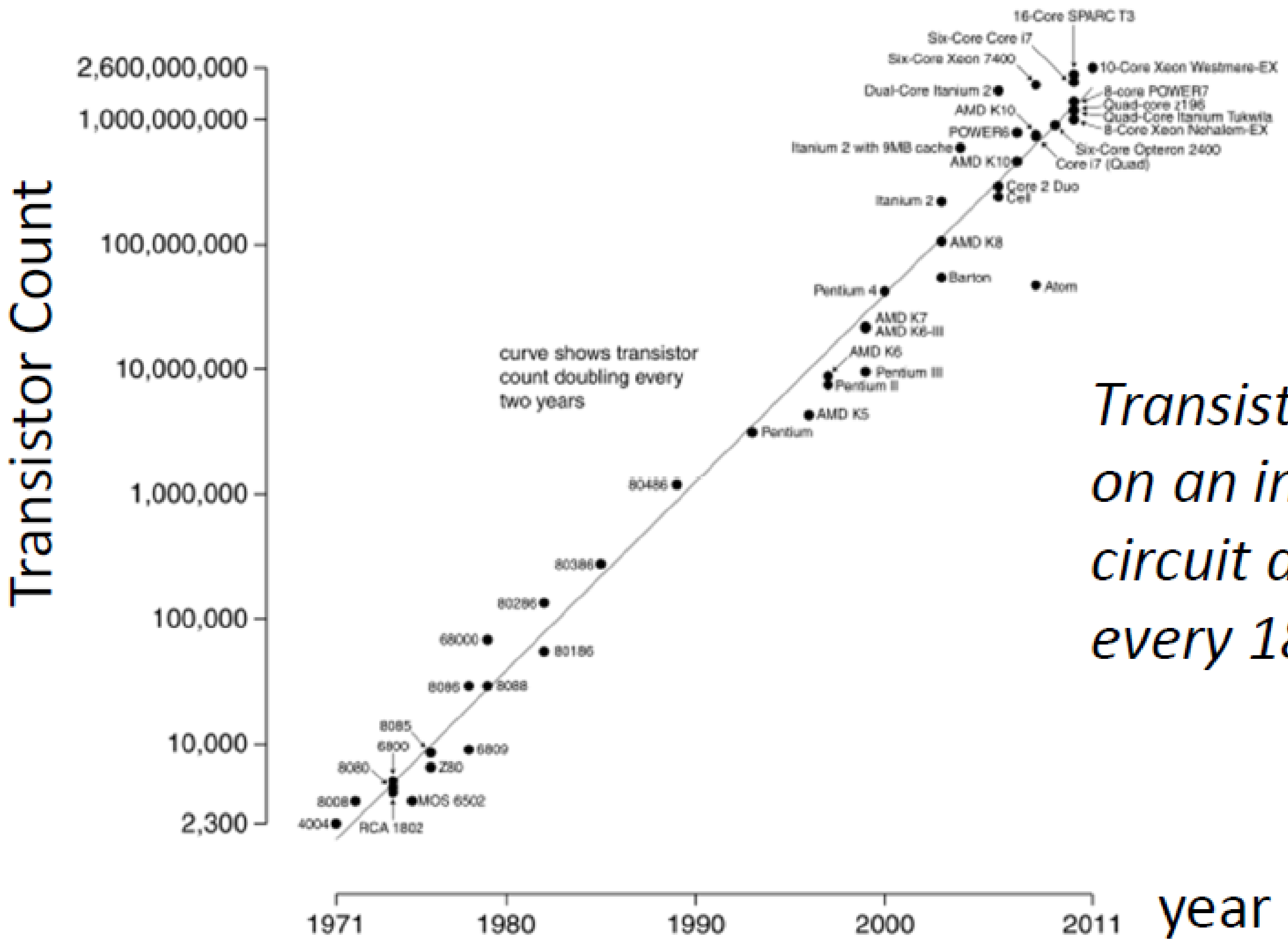
(d)

Surface-Mount Circuits

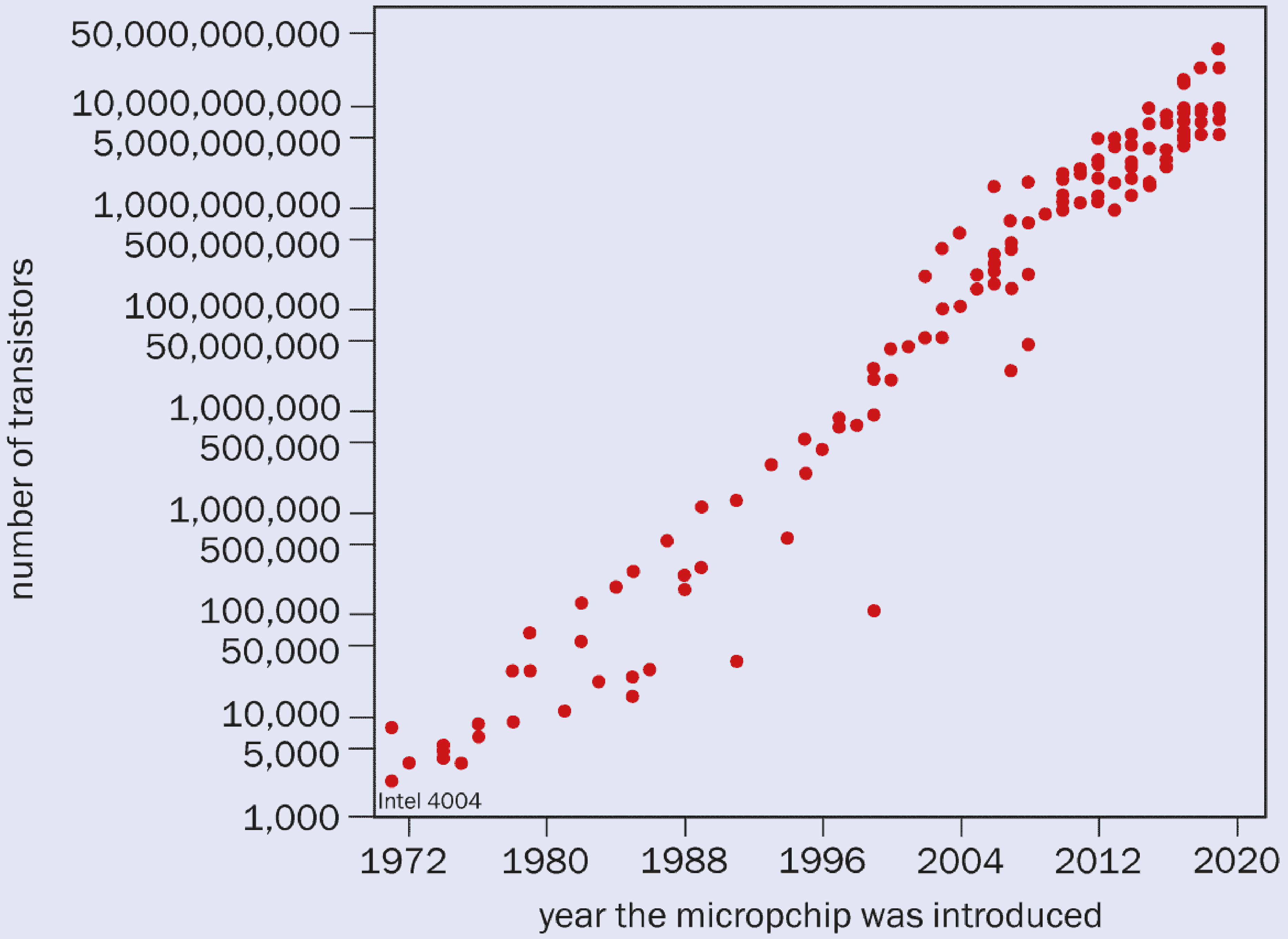
Transistors and Moore's Law

- Transistor: The fundamental building block of modern electronic devices.
- Moore's Law: the number of transistor that can be placed inexpensively on an integrated circuit die doubles every 18 months.

Transistors and Moore’s Law



Transistor count on an integrated circuit doubles every 18 months.



50,000,000,000
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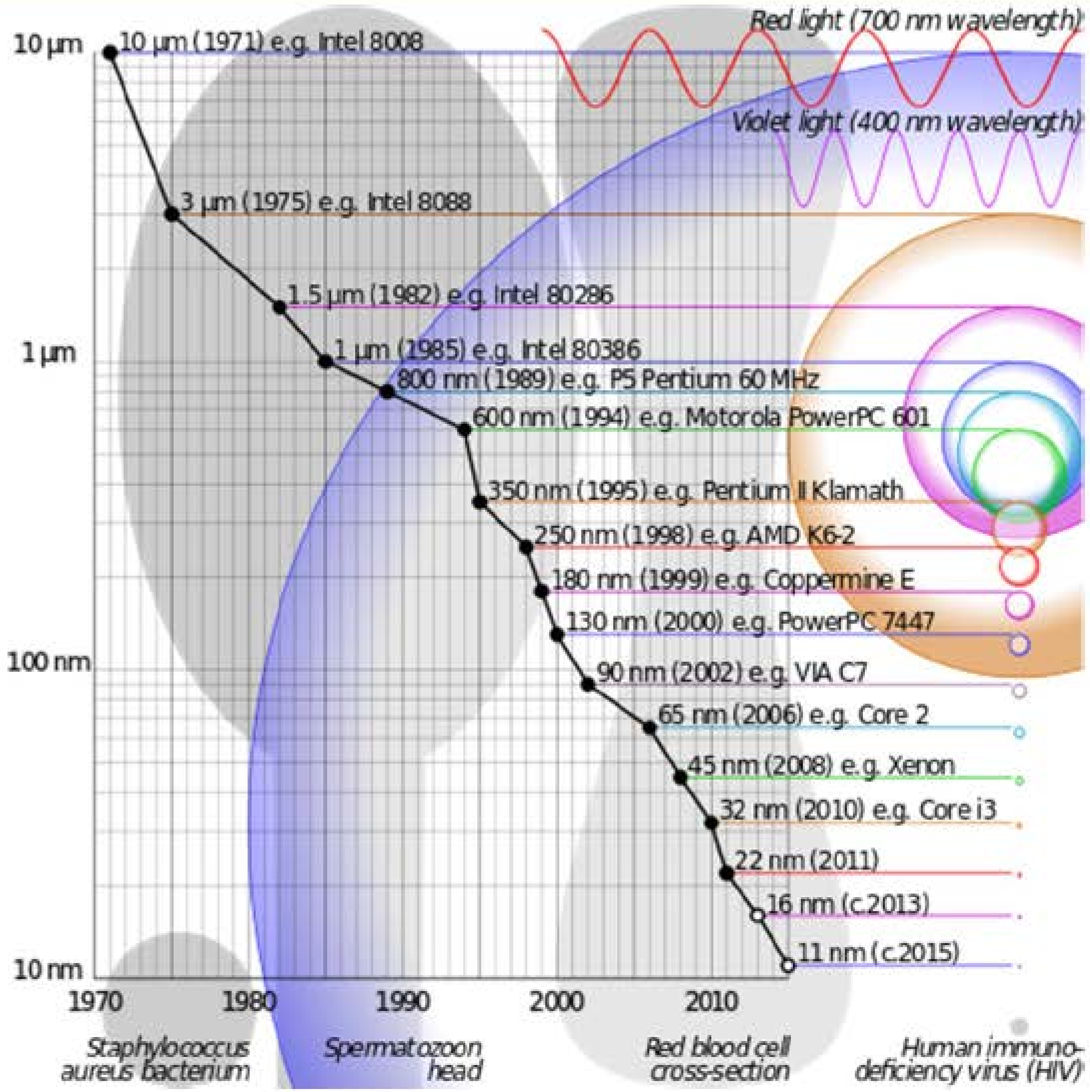
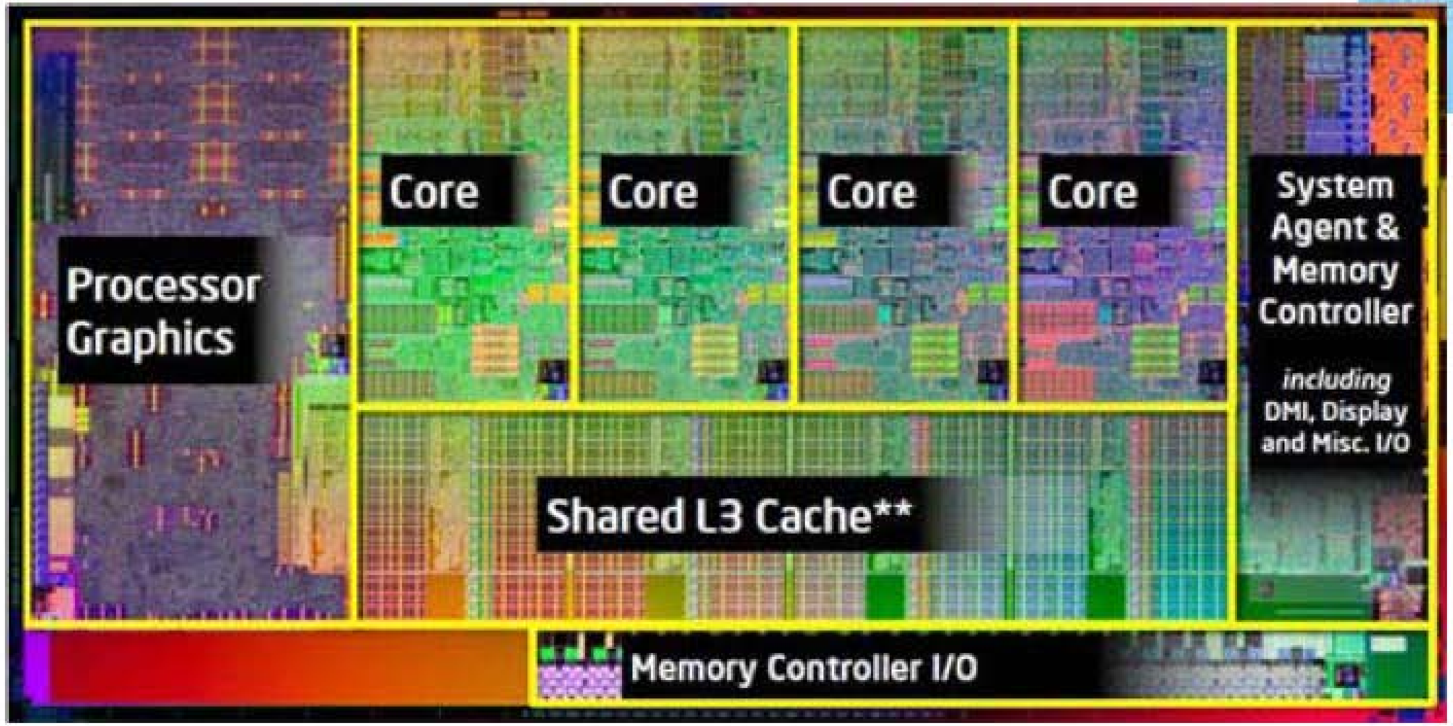
10,000
5,000
1,000
number of transistors
Intel 4004
1972
1980
1988
1996
2004
2012
2020
year the micropchip was introduced

Minimum Feature Size



32 nm technology (2010)

- Intel's Core i3, i5, i7 processors



Powerful Roadmap with Rapid Innovation

Continued Ampere Innovation Leads to Breakaway Performance Using Unique Architecture

2020 2021 2022 2023 2024 2025 2026

Ampere® Altra® Family

Up to 80 Cores
7nm

Up to 128 Cores
7nm

AmpereOne® Family

Up to 192 Cores
5nm

Up to 256 Cores
3nm

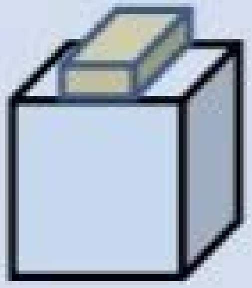
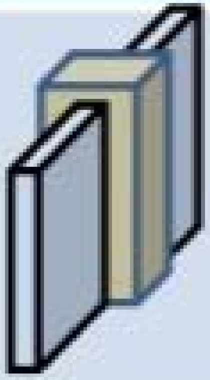
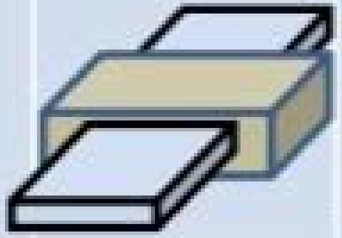
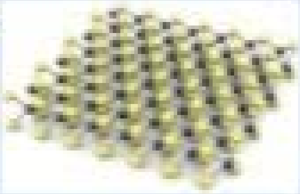
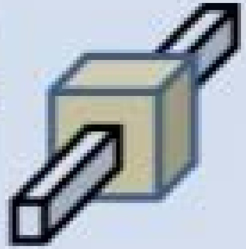
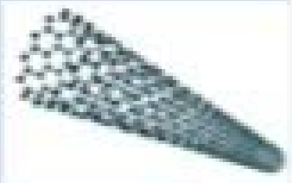
8-Channel DDR4

8-Channel DDR5

12-Channel DDR5

Generally Available

Ready at Fab

Type	Bulk	FinFET	Nano-sheet	2D-TMD	Nano-wire	CNT
Structure						
L_{MIN}^{*} (nm)	25-30	2.5-3 W_{FIN}	2.5 T_{CH}	2.5 T_{CH}	1.5 d_W	1.5 d_{CNT}

About electronics …

- Electronics is everywhere
- A field at the leading edge of technology, with rapid rate of progress
- Pushes the limits in speed, degree of integration, automation …

After this course, you'll be able to …

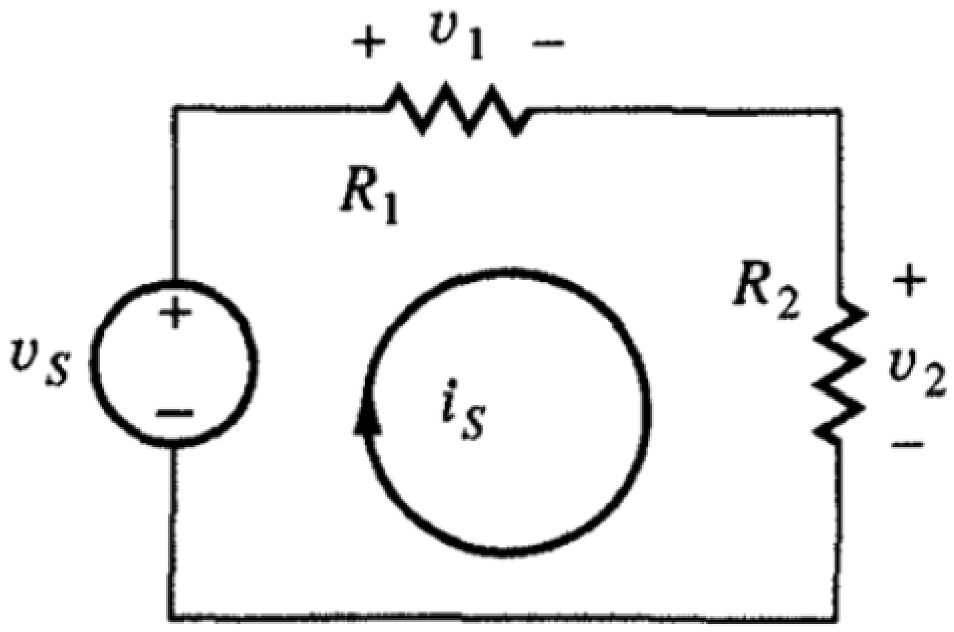
- Calculate conduction properties of materials and simple device structures
- Explain the operating principles of semiconductor diodes and FETs
- Determine the in-circuit operating state of diodes and FETs
- Perform large signal analysis of circuits containing diodes and FETs
- Use a modern schematic capture and computer-aided circuit analysis program (SPICE)
- Calculate the performance parameters for different MOS logic families to design and build circuits

Devices and Circuits I

Circuit Theory Review

- Starting point:
  - Ohm's Law (OL)
  - Kirchhoff's voltage law (KVL)
  - Kirchhoff's current law (KCL)
- Derive:
  - Voltage Division
  - Current Division
  - Thevenin Equivalent Circuits
  - Norton Equivalent Circuits

Voltage Division

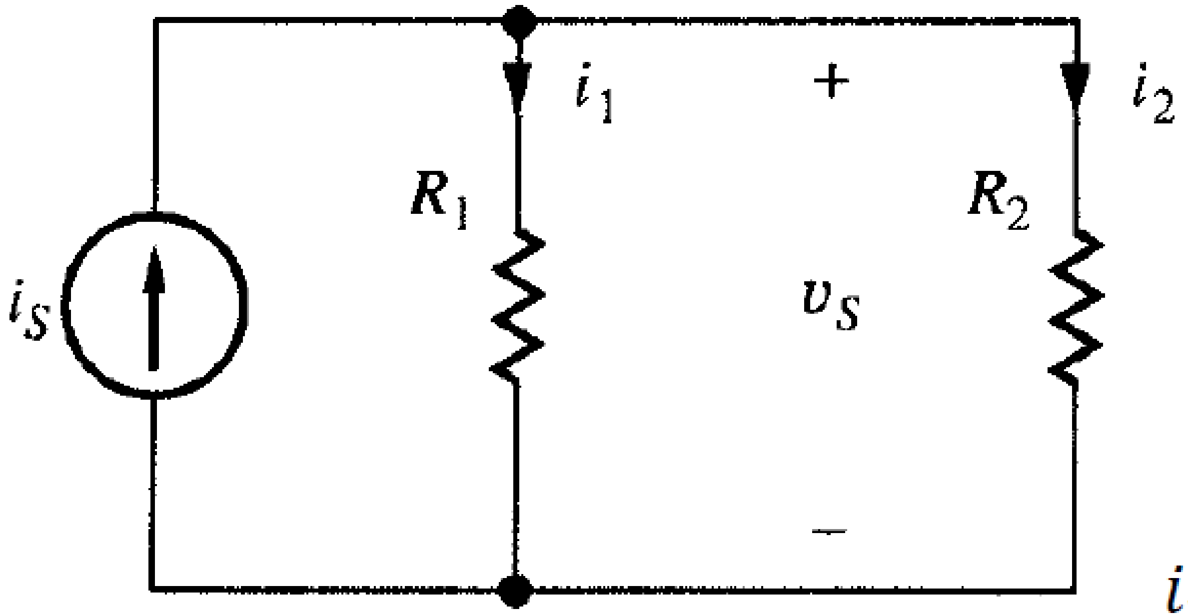


OL: $v_1 = i_S R_1$
OL: $v_2 = i_S R_1$
KVL: $v_s = v_1 + v_2$
$= i_S(R_1 + R_2)$

$$v_1 = v_s \frac{R_1}{R_1 + R_2}$$

$$v_2 = v_s \frac{R_2}{R_1 + R_2}$$

Current Division



OL: $i_1 = \frac{v_S}{R_1}$

OL: $i_2 = \frac{v_S}{R_2}$

KCL: $i_s = i_1 + i_2$

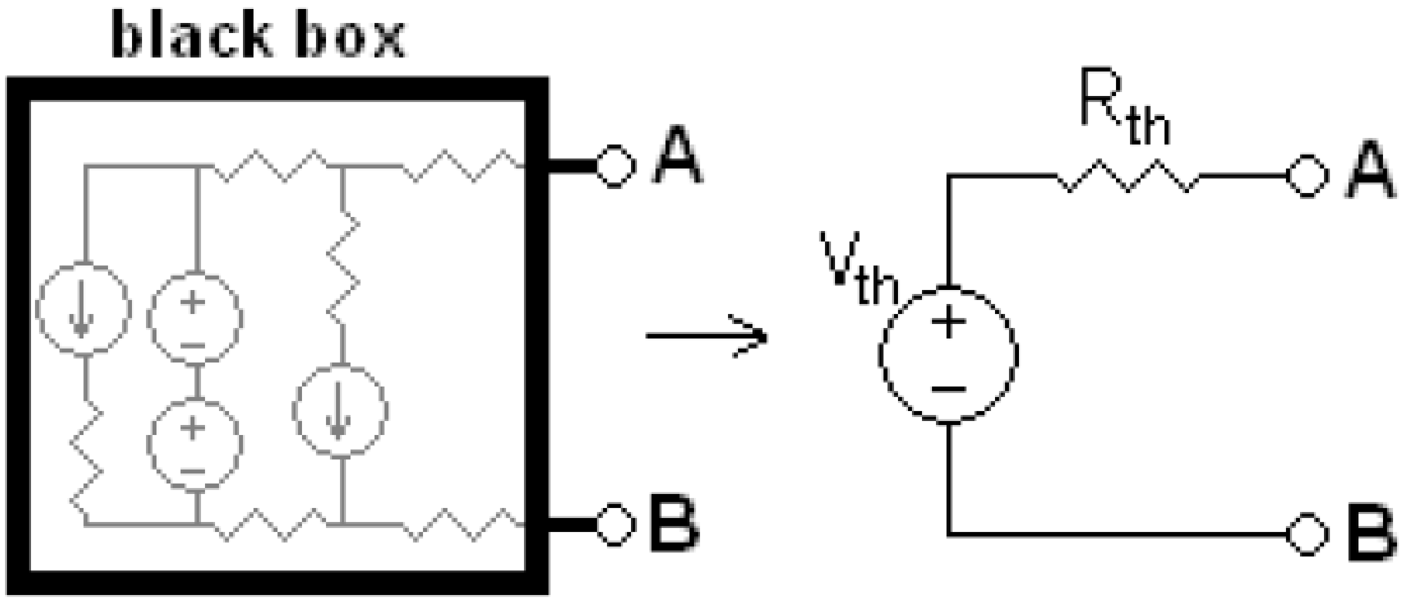
$= \frac{v_S}{R_1} + \frac{v_S}{R_2} = \frac{v_S}{(R_1||R_2)}$

$$i_1 = i_s \frac{R_1||R_2}{R_1} = i_s \frac{R_2}{R_1 + R_2}$$

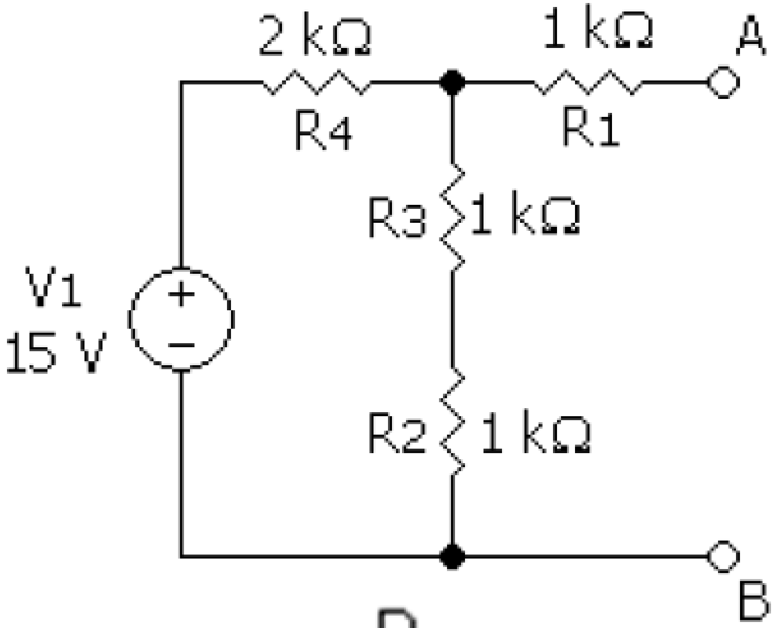
$$i_2 = i_s \frac{R_1||R_2}{R_2} = i_s \frac{R_1}{R_1 + R_2}$$

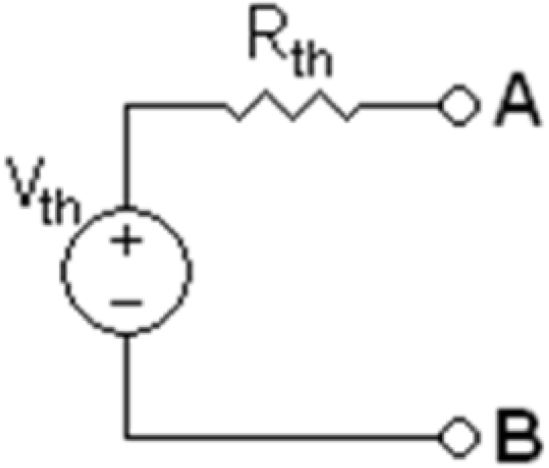
Thevenin Equivalent Circuits

- Voltage source V_{Th} is the open circuit voltage at the output terminals
- R_{Th} : equivalent resistance present at the output terminals with all independent sources set to zero



Thevenin Equivalent Circuits (e.g. 1)





Open circuit voltage:

$$V_{Th} = V_{23}$$

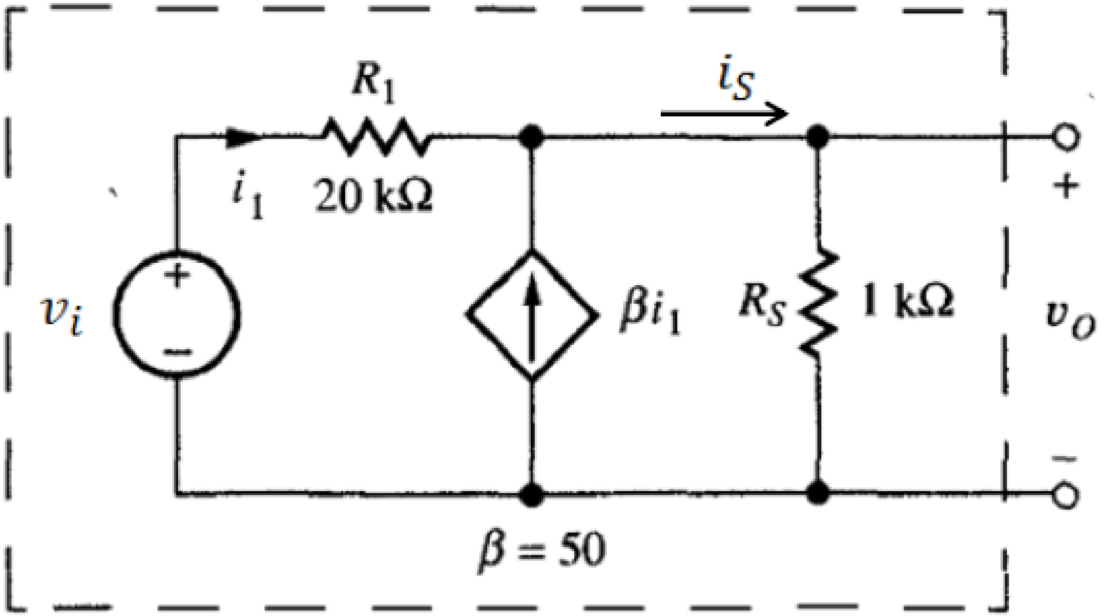
voltage division

$$= V_1 \frac{R_4}{R_4 + R_2 + R_3} = 7.5\ \mathrm{V}$$

Set V_1 to zero.

$$R_{Th} = R_1 + R_4 || (R_2 + R_3)$$
$$= 1\ \mathrm{k\Omega} + 2\ \mathrm{k\Omega} || 2\ \mathrm{k\Omega} = 2\ \mathrm{k\Omega}$$

Thevenin Equivalent Circuits (e.g. 2)



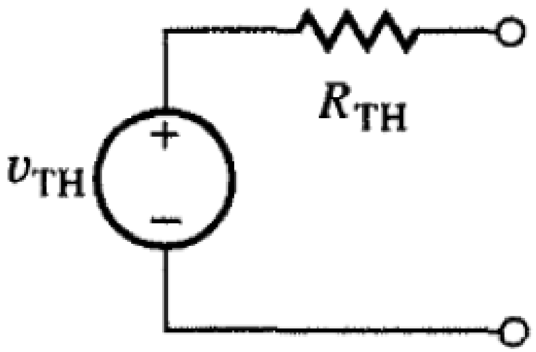
$$V_{Th} = i_S R_S$$

KCL: $i_S = (1+\beta)i_1$

$$\Rightarrow V_{Th} = (1+\beta)i_1 R_S$$

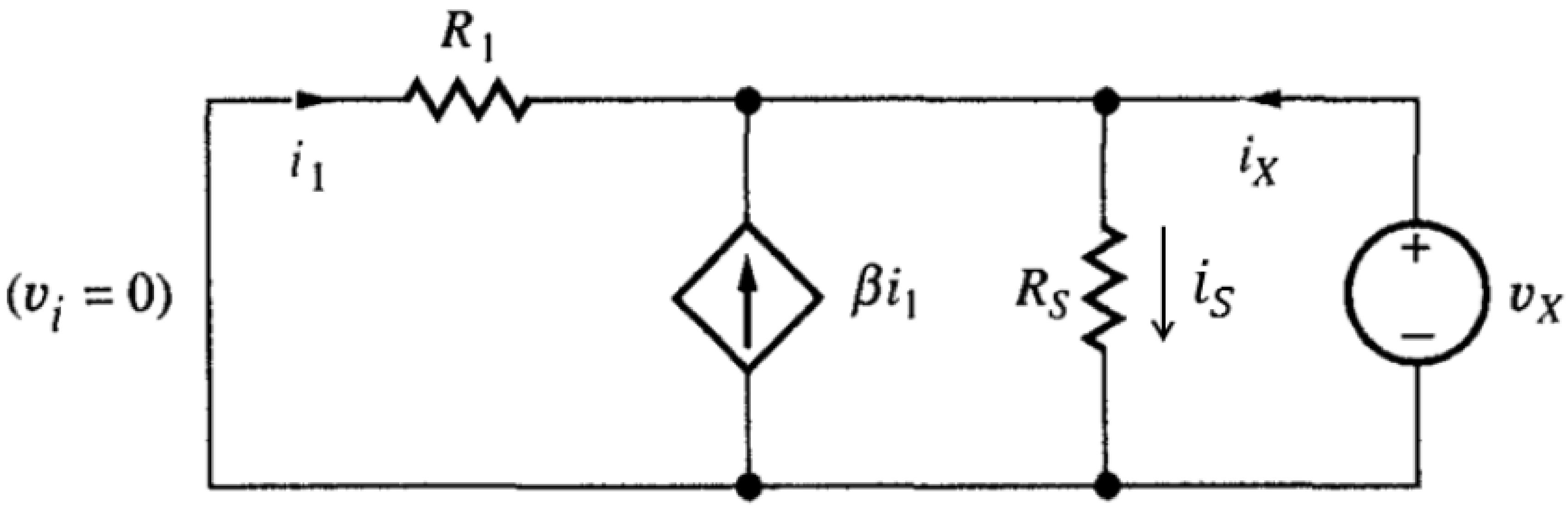
KVL:

$$v_i = i_1 R_1 + (1+\beta)i_1 R_S$$



$$V_{Th} = v_S \frac{(1+\beta)R_S}{R_1 + (1+\beta)R_S}$$

Thevenin Equivalent Circuits (e.g. 2)



KVL: $v_X = -i_1 R_1$; $v_X = i_S R_S$

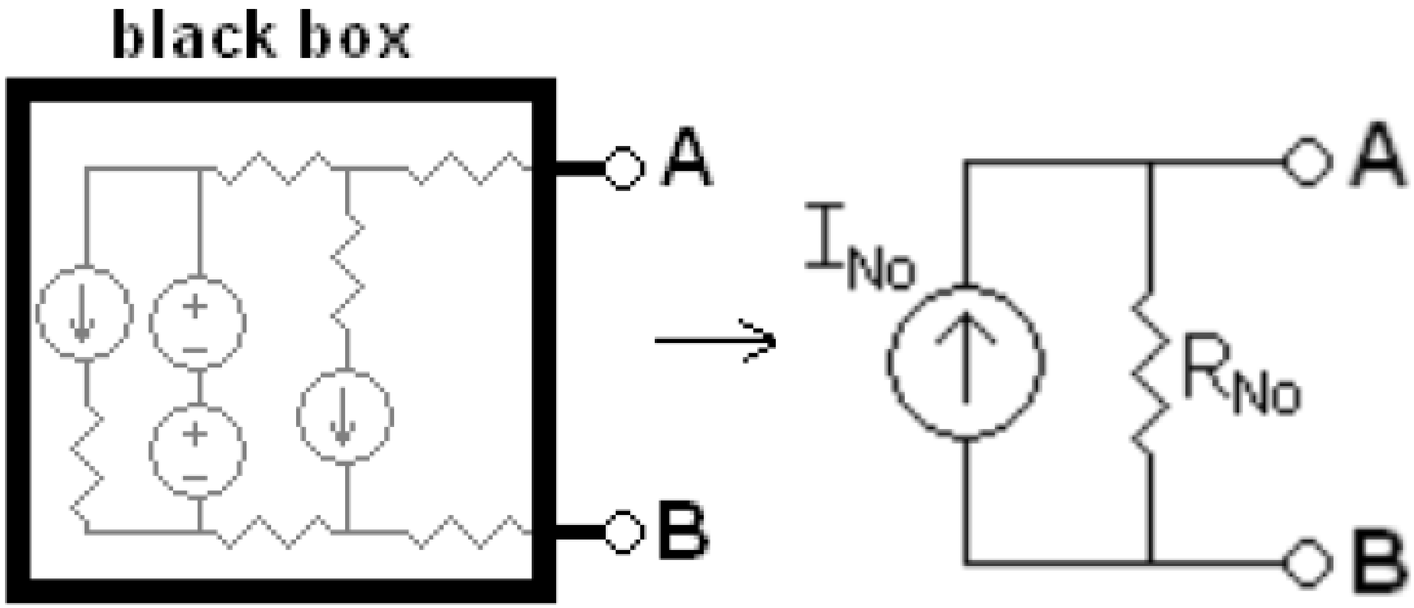
KCL: $i_S = i_X + (1+\beta)i_1$

$\Rightarrow v_X = [i_X + (1+\beta)i_1]R_S = \left[i_X - (1+\beta)\frac{v_X}{R_1}\right]R_S$

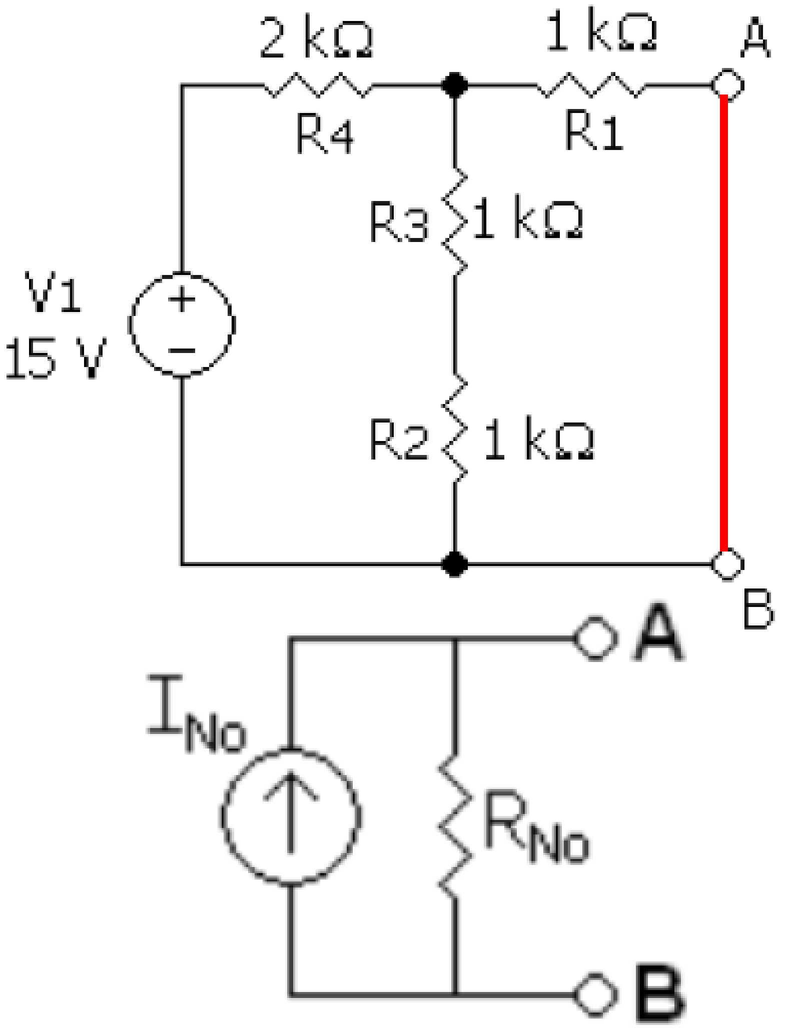
$\Rightarrow R_{Th} = \frac{v_X}{i_X} = v_S \frac{R_S}{1+(1+\beta)R_S/R_1}$

Norton Equivalent Circuits

- I_{No}: current coming out of the network when terminals are shorted
- R_{No} : equivalent resistance present at the output terminals with all independent sources set to zero ($R_{No} = R_{Th}$)

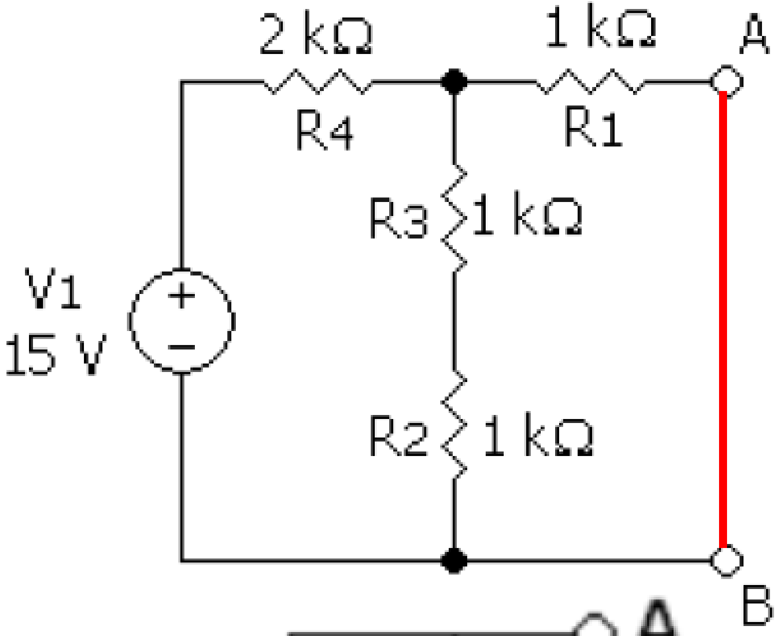


Norton Equiva



Find Norton equivalent In and Rn

Norton Equivalent Circuits (e.g. 1)



$$i_{tot} = \frac{V_1}{R_{tot}} = \frac{V_1}{R_4 + R_1||(R_2 + R_3)} = 5.63 \text{ mA}$$

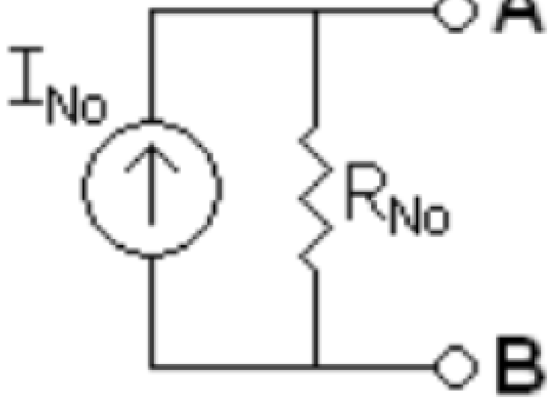
Short Circuit current:

current division

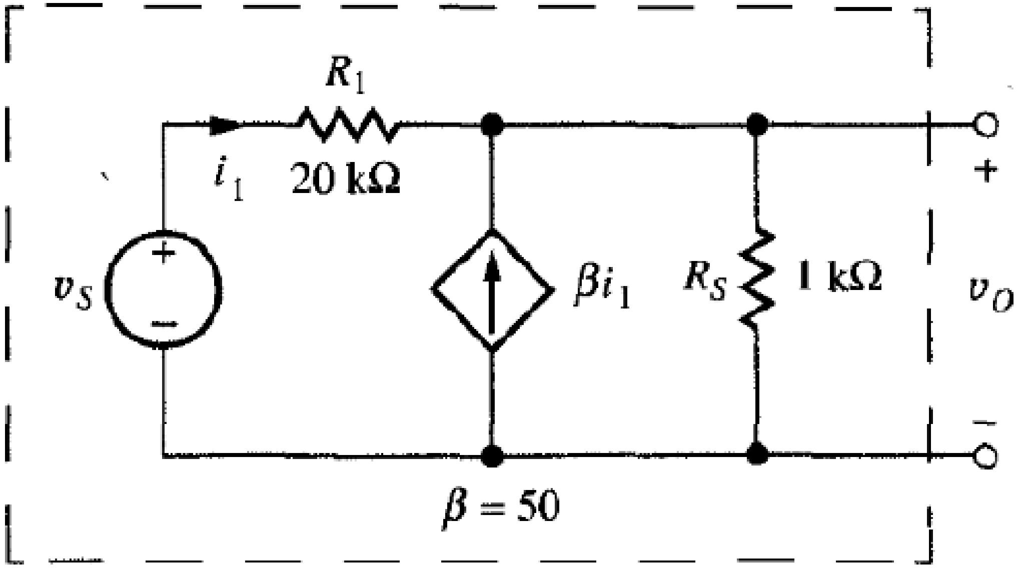
$$I_{No} = i_{tot}\frac{R_2 + R_3}{R_2 + R_3 + R_4} = 3.75 \text{ mA}$$

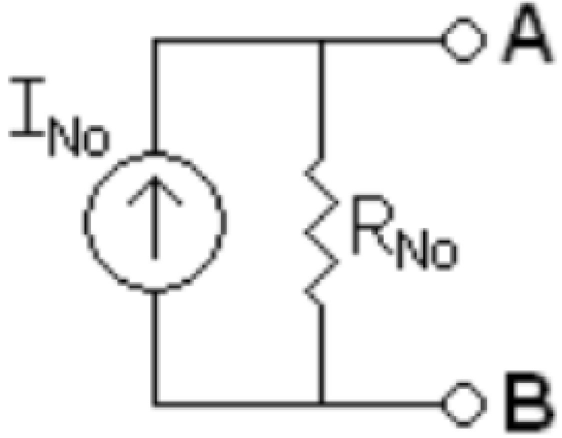
Set V_1 to zero.

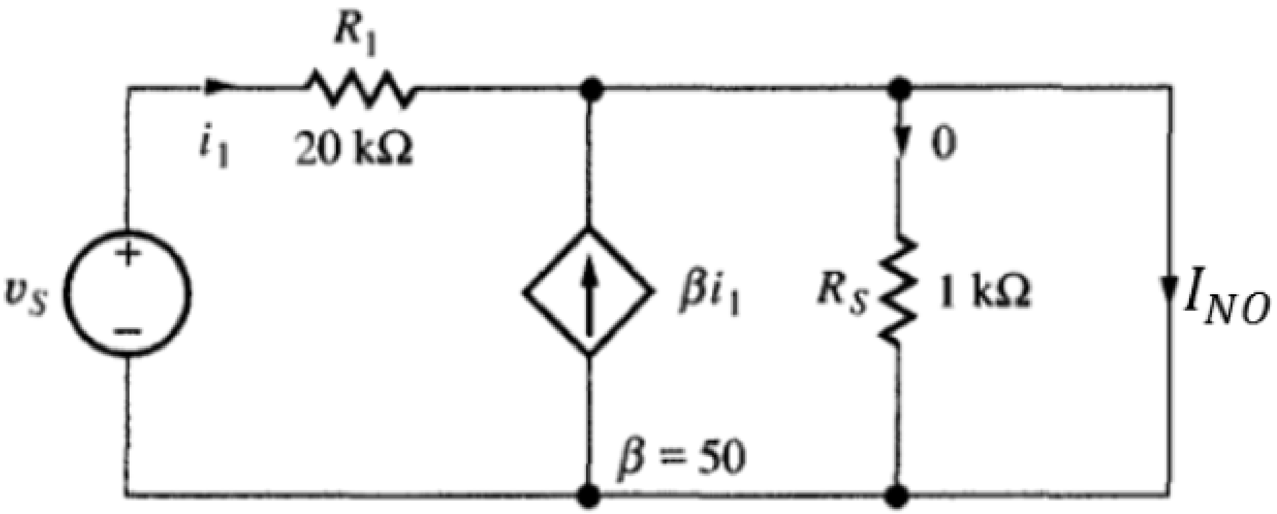
$$R_{No} = R_1 + R_4||(R_2 + R_3)$$
$$= 1 \text{ k}\Omega + 2 \text{ k}\Omega||2 \text{ k}\Omega = 2 \text{ k}\Omega$$



Norton Equivalent Circuits (e.g. 2)







$$I_{No} = (1+\beta)i_1$$
$$= (1+\beta)v_s/R_1$$

R_{No} is the same as R_{Th} in Thevenin equivalent circuits

$$V_{Th} = I_{No}R_{Th}$$